

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Implemented backend security controls, **AWS KMS** encryption for PII/clinical data, 90-day retention, auth hardening, and rate limiting, and remediated **20+ VAPT findings**.
- Served as primary technical POC for the eTriage backend; ran knowledge transfer onboarding a second developer to on-call readiness in **~6 weeks** and piloted a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production open-source LLM auto-commenting feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN gym-management platform with Stripe + manual-cash hybrid payments, cron automation, and multi-branch support; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests**.

Refusal Decay – LLM Safety / Mechanistic Interpretability (GitHub) PyTorch, Llama-3.1-8B, AdvBench

- Studied how prefilling attacks weaken refusal in Llama-3.1-8B-Instruct, extracting the residual-stream refusal direction via difference-in-means (refusal **0.92**→**0.32** at $k=3$) with causal patching, bootstrap 95% CIs, and McNemar's test.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, C++, Java, SQL, Bash, MongoDB, PostgreSQL, SQLite
Frameworks / Tools: FastAPI, Node.js, Express, React, Docker, Git, GitHub Actions, Cloudflare Workers, REST APIs
ML & AI: PyTorch, JAX, Hugging Face Transformers, sentence-transformers, LLM agents, RAG, MCP, evaluation
Retrieval / Search: RRF, cross-encoder reranking, MMR, knapsack token budgeting, Brave Search, cell-level provenance
Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.